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Hominoid visual brain structure volumes and the position of the lunate sulcus

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ABSTRACT

It has been argued that changes in the relative sizes of visual system structures predated an increase in brain size and provide evidence of brain reorganization in hominins. However, data about the volume and anatomical limits of visual brain structures in the extant taxa phylogenetically closest to humans—the apes—remain scarce, thus complicating tests of hypotheses about evolutionary changes. Here, we analyze new volumetric data for the primary visual cortex and the lateral geniculate nucleus to determine whether or not the human brain departs from allometrically-expected patterns of brain organization. Primary visual cortex volumes were compared to lunate sulcus position in apes to investigate whether or not inferences about brain reorganization made from fossil hominin endocasts are reliable in this context. In contrast to previous studies, in which all species were relatively poorly sampled, the current study attempted to evaluate the degree of intraspecific variability by including numerous hominoid individuals (particularly *Pan troglodytes* and *Homo sapiens*). In addition, we present and compare volumetric data from three new hominoid species—*Pan paniscus*, *Pongo pygmaeus*, and *Symphalangus syndactylus*. These new data demonstrate that hominoid visual brain structure volumes vary more than previously appreciated. In addition, humans have relatively reduced primary visual cortex and lateral geniculate nucleus volumes as compared to allometric predictions from other hominoids. These results suggest that inferences about the position of the lunate sulcus on fossil endocasts may provide information about brain organization.

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Introduction

The primary visual cortex (V1) receives visual information directly from the lateral geniculate nucleus (LGN) in the thalamus, which in turn receives information from the retinal ganglion cells via the optic nerve. In humans, V1 appears to be smaller than would be predicted for a primate of our brain size (Filimonoff, 1933; Frahm et al., 1984; Holloway, 1997), and LGN is smaller than predicted for a primate of our brain size (Holloway, 1997). It has been suggested that this pattern reflects an increase in brain tissue allocated to higher order functions, and does not reflect a reduction in visual information processing (Holloway, 1997), because comparable visual acuity has been demonstrated for humans and macaques (Macaca; De Valois et al., 1974). In fact,

the absolute volume of V1 in humans exceeds that of all other primates (Frahm et al., 1984; Bush and Allman, 2004), although it does not keep pace with the three-fold expansion of human neocortex over that of great apes.

In fossil hominins, evidence for changes in V1 volume have been inferred from the position of the lunate sulcus (also called the Affenspalte or simian sulcus), a gross anatomical landmark approximately coincident with the lateral-anterior limit of V1 in apes and some monkey species (Fig. 1; von Bonin and Bailey, 1947; Holloway et al., 2003a). Dart (1925) observed a posteriorly-positioned lunate sulcus in the Taung (*Australopithecus africanus*) endocast, and suggested that this indicates enlargement of posterior parietal association cortex at the expense of V1 volume. Posterior parietal areas have specialized multisensory and motor functions in behaviors which include gesturing (Creem-Regehr, 2009), action planning (Coulthard et al., 2008), tool use (Peeters et al., 2009), and stone tool making (Stout et al., 2008). In contrast, V1 is entirely visual in function and relatively unspecialized.

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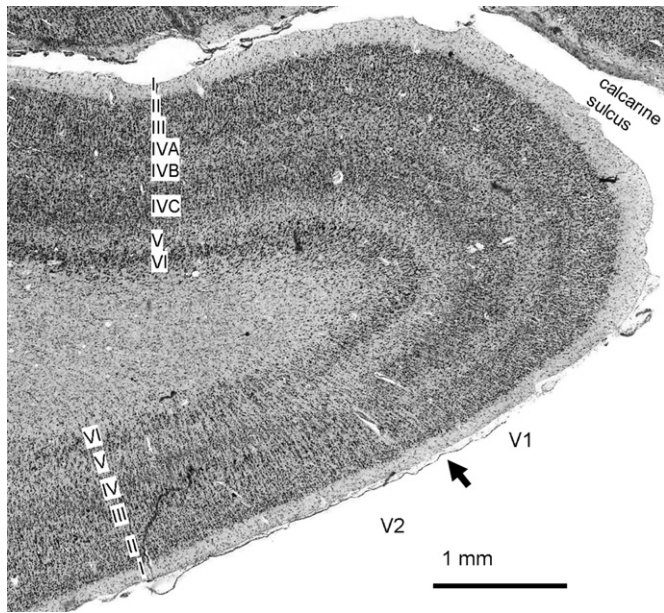


Fig. 1. Coronal section of *Gorilla gorilla* brain near the calcarine sulcus indicating the border between V1 and V2 (at arrow). Note the differences in laminar pattern between V1 and V2. For example, in V1 layer 4 is subdivided into sublayers 4A, 4B and 4C. A band of fibers, called the stripe of Gennari, passes through layer 4B, which is sparse in cell bodies on Nissl sections. Also note large pyramidal cells in deep layer 3 of V2, immediately adjacent to the border with V1.

Although Dart's interpretation of the lunate sulcus in Taung has been questioned (Falk, 1980), another endocast, Stw 505, has been proposed to provide better evidence for a posteriorly-positioned lunate sulcus in *Australopithecus africanus* (Holloway et al., 2004b). Holloway (1966, 1975, 1985) has discussed in detail the notion that brain reorganization might have enabled small-brained early hominins to engage in more complex behaviors.

There are two potential problems with relying on the lunate sulcus as an indication of brain organization. First, although sulci are often used as landmarks for determining the size of cortical regions in hominoids, they do not reliably delimit cytoarchitectonic areas (Amunts et al., 2007b). In fact, correspondence between histologically-defined V1 volume and the extent of the lunate sulcus has not been demonstrated in apes. However, current evidence shows that the position of the lunate sulcus indicates the extent of the lateral part of V1 in chimpanzees (*Pan troglodytes*; including two specimens with lunate sulci in unusually posterior positions; Holloway et al., 2003b). In fact, Smith (1903) and Black (1915) considered a relationship to striate cortex to be a requirement for lunate sulcus identification in humans, based on the observation that the lunate sulcus delimits V1 in the species in which it occurs (Brodman, 1906; Ingalls, 1914). However, it is possible that the lunate sulcus is only an indication of the lateral-anterior limit of V1, and does not correlate with total V1 volume in closely related hominoid species. In humans Ono et al. (1990) report that a lunate sulcus is present in 60% of right and 64% of left hemispheres ($n = 25$), although Allen et al. (2006) argue that "true" lunate sulci are much rarer, only occurring in 1.4% of human hemispheres ($n = 220$). In humans, the sulcus best associated with the extent of V1 is the calcarine sulcus, which is located on the medial surface of the hemispheres. However, Gilissen et al. (1995) suggest that V1 volume cannot be inferred from calcarine sulcus length in human brains ($n = 23$) because the depth of the sulcus varies considerably.

Second, it is possible that reduced relative V1 volume can be attributed to increased brain size, unrelated to functionally-relevant changes in brain organization. Across primates, V1 volume scales with negative allometry to brain size (Frahm et al., 1984; Bush and Allman, 2004), so it is possible that reduction in relative V1 volume in humans is simply an extension of this allometric relationship. It has also been suggested that the position of the lunate sulcus is directly related to brain size—in bigger brained species, the extent of V1 is shifted from a more-lateral to a more-medial position—and therefore, the lunate sulcus cannot occur in a posterior position on a small-brained hominoid (Jerison, 1975).

Several hypotheses have been proposed to explain observations about the negative allometric scaling of V1 volume to overall brain volume. Kaas (2000) hypothesized that, as cortical areas increase in surface area, it becomes more difficult to maintain connections among them, and as a result, the number of cortical areas increases. This is supported by the finding that across mammals the number of neocortical areas (and the number of areas to which each is connected) scale to the $1/3$ power of the volume of the cerebral cortex (Changizi and Shimojo, 2005). Thus, as a general rule, larger brains tend to have a greater number of visual areas. The role of the primary cortical area in information processing is expected to decrease as its specific functions are delegated to an increasing number of higher order cortical areas. It follows that the relatively large macaque V1 would be more generalized in function than the relatively small human V1. Although such physiological differences have not yet been demonstrated, histological differences between human and macaque V1 have been established (Preuss et al., 1999; Preuss and Coleman, 2002).

We measured the volume of V1 and other brain structures in hominoids and a crab-eating macaque (*Macaca fascicularis*) to test the hypothesis that the human V1 volume can be predicted from allometrically-expected patterns of brain organization. Additionally, nonhuman hominoid brains were three-dimensionally reconstructed such that V1 volumes could be compared to the position of the lunate sulcus. In particular, this study sets out to examine whether or not previous findings that humans have significantly reduced V1 and LGN volumes can be confirmed for a larger comparative sample of hominoid brains, and whether or not the lunate sulcus serves as a reliable predictor of V1 volume in nonhuman hominoid species. Using these new comparative visual structure volume data from hominoids, we further examine the suggestion that humans differ from nonhuman hominoid species in the relative size of specific brain regions, but that these differences are small compared to differences in visual system structures (Holloway, 1997, 2002).

Materials and methods

Specimens and tissue preparation

Measurements were taken on histological sections from a total of 29 brains representing seven hominoid species, plus one cercopithecoid species. We used sections from the left hemispheres of adult specimens, including a range of ages and both sexes (Table 1). To accumulate a large and diverse sample, specimens in the study come from several different collections: the Zilles and Stephan comparative neuroanatomy collections at C&O Vogt Institute of Brain Research in Düsseldorf, Germany, the Yakovlev-Haleem and Welker collections at the Armed Forces Institute of Pathology in Washington, D.C., and the Great Ape Aging Project at the George Washington University, Washington, D.C. and the Mount Sinai School of Medicine, New York, N.Y.

The human and nonhuman hominoid brains from the Zilles collection were immersion fixed with either 4% formaldehyde or

Table 1
Specimens and volumes

species	code	collection	brain mass (g)	correction factor	brain vol. (cm ³)	left V1 vol. (cm ³)	left LGN vol. (cm ³)	neocortex vol. (cm ³) ^a
<i>Homo sapiens</i> ^b	14686	Zilles	1437	2.01	1387.1	9.4	0.2	1116.8
<i>Homo sapiens</i> ^b	1696	Zilles	1757	2.44	1695.9	10.4	0.3	NA
<i>Homo sapiens</i> ^b	18992	Zilles	1270	2.19	1225.9	7.5	0.2	NA
<i>Homo sapiens</i> ^b	20784	Zilles	1349	2.14	1302.1	8.5	0.2	NA
<i>Homo sapiens</i> ^b	28193	Zilles	1360	2.12	1312.7	9.2	0.2	NA
<i>Homo sapiens</i> ^b	295	Zilles	1046	1.66	1009.7	5.3	0.1	NA
<i>Homo sapiens</i> ^b	SN382/81	Zilles	1142	2.05	1102.3	6.0	0.1	865.2
<i>Homo sapiens</i> ^b	54491	Zilles	1350	1.92	1303.1	7.6	0.2	NA
<i>Homo sapiens</i> ^b	5694	Zilles	1216	1.81	1173.7	5.4	0.1	NA
<i>Homo sapiens</i> ^b	6895	Zilles	1110	1.46	1071.4	7.0	0.2	NA
<i>Pan troglodytes</i>	1548	Zilles	NA	2.05 ^c	265.0	2.8	0.1	198.3
<i>Pan troglodytes</i>	Bathsheba 4/97	Zilles	359.5	2.00	347.0	4.7	0.2	262.8
<i>Pan troglodytes</i>	Schimp 1	Zilles	440	1.99	424.7	4.8	0.2	297.4
<i>Pan troglodytes</i>	YN89–278	Zilles	420	2.24	405.4	5.2	0.2	300.1
<i>Pan troglodytes</i>	3.1.1.0.1	Yakovlev-Haleem	NA	2.05 ^c	261.1	4.0	0.1	NA
<i>Pan troglodytes</i>	3.1.1.0.3	Yakovlev-Haleem	NA	2.05 ^c	276.8	3.8	0.1	NA
<i>Pan troglodytes</i>	56_49	Welker	NA	2.05 ^c	257.7	3.7	NA	NA
<i>Pan paniscus</i>	YN86–137	Zilles	392	2.60	378.4	7.3	0.2	279.0
<i>Pan paniscus</i>	Zahlia 1/97	Zilles	324	1.63	312.7	5.7	0.1	214.4
<i>Pan paniscus</i>	MCZ-2007-52	GWU	337	2.74	325.3	4.4	0.1	NA
<i>Gorilla gorilla</i>	YN82–140	Zilles	376	2.04	362.9	4.0	0.2	254.3
<i>Gorilla gorilla</i>	A375	Stephan	450	1.87	434.4	5.1	0.2	313.1
<i>Pongo pygmaeus</i>	Harry 2/97	Zilles	440	2.28	424.7	4.2	0.1	298.9
<i>Pongo pygmaeus</i>	YN85–38	Zilles	369	2.08	356.2	3.5	0.1	268.5
<i>Pongo pygmaeus</i>	Briggs 5/97	Zilles	345	2.15	333.0	4.7	0.2	240.3
<i>Hylobates lar</i>	YN81–146	Zilles	92	2.09	88.8	1.8	0.1	59.6
<i>Hylobates lar</i>	Disco 3/97	Zilles	120	2.16	115.8	2.3	0.1	77.0
<i>Symphalagus syndactylus</i>	880804	GWU	138.7	2.91	133.9	2.7	NA	NA
<i>Macaca fascicularis</i>	ma22	Zilles	57.6	1.90	55.6	1.4	0.0	NA

^a Neocortex volume includes grey matter and underlying white matter^b These *Homo sapiens* brain and left V1 volumes differ slightly from those in a previous publication (Amunts et al., 2007a) in which they were shrinkage-corrected based on the an estimated specific gravity of brain tissue of 1.032 g/cm³ (Zilles, 1972) rather than 1.036 g/cm³ (Stephan, 1960).^c *Pan troglodytes* mean correction factor used because brain weight unknown.

Bodian's solution within a few hours after death, embedded in paraffin and serially-sectioned along the coronal plane at a thickness of 20 µm (except for one chimpanzee brain, which was horizontally sectioned at a thickness of 15 µm), and stained for cell bodies based on Gallyas' procedure (Gallyas, 1971), using silver (Ag) according to the technique described by Merker (1983). The *Macaca fascicularis* brain from the Zilles collection was perfusion-fixed with 4% formaldehyde in phosphate buffer, embedded in paraffin and serially-sectioned along a coronal plane at 20 µm, and Merker-stained for cell bodies. The *Gorilla gorilla* brain from the Stephan collection was perfused *in situ* with Bouin's fluid through the carotid arteries after the blood was washed out with physiological saline, embedded in paraffin and serially-sectioned along the coronal plane at a thickness of 20 µm, and Nissl-stained using cresyl violet. The *Pan troglodytes* brains from the Yakovlev-Haleem collection were sagittally-sectioned at a thickness of 35 µm, and separate alternating series were stained for Nissl and for myelin. The *Pan troglodytes* brain (Welker collection) was coronally sectioned at a thickness of 50 µm and Nissl-stained using thionin. Hominoid brains housed at the George Washington University (GWU) were immersion fixed in 10% neutral buffered formalin for no more than 14 days. Left occipital lobe and parieto-occipital lobe blocks were cryoprotected by immersion with increasing concentrations of sucrose solutions up to 30%, frozen on dry ice, and serially-sectioned on a microtome at a thickness 40 µm, and Nissl-stained with cresyl violet. For further details about the sections (see Supplemental Online Materials, Table 1). Multiple collections were used to include as large a sample as possible, but it is prudent to keep methodological considerations in mind as the

tissue preparation techniques varied across collections, which may impact volume measurements (Stephan et al., 1981).

Magnetic resonance image acquisition and distance measures

Magnetic resonance images were used to make 3D brain reconstructions from which gross anatomical landmarks could be identified, and the distances between them measured and compared to volumetric data. These were available for *Pan troglodytes* ($n = 1$), *Pan paniscus* ($n = 1$), *Pongo pygmaeus* ($n = 2$) and *Hylobates lar* ($n = 1$) specimens from the Zilles collection which had also been measured for V1 volumes. MR imaging of postmortem brains, was carried out as previously described and validated (Roland and Zilles, 1994; Zilles et al., 1995; Amunts et al., 1996; Geyer et al., 1996). Imaging was performed with a Siemens 1.5-T magnet (Erlangen, Germany) with a T1-weighted 3-D FLASH sequence covering the entire brain (flip angle 40°, repetition time TR = 40 ms, echo time TE = 5 ms for each image). Each volume consisted of 128 sagittal sections. The spatial resolution was 0.75 × 0.75 × 0.75 mm. The image space was divided into voxels each with a resolution of 8 bits corresponding to 256 gray values.

To improve visualization of the cerebral cortex, the brainstem and cerebellum were segmented from the brain, and the left and right hemispheres were segmented from each other using in-house software (HNV, developed by Hartmut Mohlberg, Jülich Research Center Institute of Medicine, Germany). 3D surface reconstructions that could be manipulated in virtual space were created using the marching cubes algorithm in Amira-Advanced 3D Visualization and Volume Modeling Package (Indeed, Visual Concepts GmbH, Berlin, Germany). Following the descriptions of Holloway (1984,

1988) and Holloway et al. (2004a), three landmarks were identified on the surfaces of the hemispheres: the occipital pole (OP, first landmark), the frontal pole (FP, second landmark), and the superior end of the lunate (LS, third landmark). The three dimensional coordinates of landmarks were localized in HMV and used to calculate linear chord distances (see [Supplementary Online Materials](#) for further details).

Estimation of volumes

Neocortex volumes were obtained as previously reported (Stephan et al., 1981). They included the grey and white matter of the cerebral cortex; the grey matter of the paleocortex (entorhinal cortex, schizocortex, hippocampus and amygdala) was segmented out. V1 volumes included the grey matter only, identified on the basis of its topological location and distinct appearance including the sub-stratification of layer 4 into multiple sublayers, a feature which is present in all anthropoids (Allman and McGuinness, 1988). The LGN volumes comprise the entire structure including all dorsal lateral geniculate nucleus laminae and interlaminar space identifiable on the Nissl-stained sections, as outlined in primate brain atlases (e.g., Stephan et al., 1980; Paxinos et al., 2000; Fig. 2). Manual outlining in Image J (Rasband, 1997–2007) was used to calculate area measurements from 600–2400 DPI scans of the histological sections. Using these areas, left V1 (cortex only), left LGN (entire structure) and whole brain (entire structure) volumes were estimated using the Cavalieri principle (Zilles et al., 1982).

For each brain, the volumes obtained from histological slides were corrected to more closely approximate *in vivo* volumes and to allow inter-individual comparison by multiplying to a shrinkage correction factor. Correction factors were obtained by dividing the fresh brain volume by the brain volume determined from microscope slides. The brain volumes were estimated from the fresh brain mass and given the average specific gravity of brain tissue (1.036 g/cm³; Stephan, 1960). These were previously reported for most specimens in the Zilles and Stephan collections (MacLeod et al., 2003).

As many specimens as possible were included, which meant including specimens that were processed through different methods. In general, Nissl-stained sections were used for making measurements, although rarely, adjacent serial sections stained for myelin, or MR sections, were used when Nissl-stained sections were missing or inadequate (for details see [Supplementary Online Materials, Table 1](#)). To address potential variability due to processing, shrinkage correction factors were calculated for each brain series, except for some *Pan troglodytes* specimens for which brain mass data were unavailable and the species' mean correction factor was used (see [Supplementary Online Materials, Table 1](#)).

Statistical analyses

All statistical analyses were assessed with an α level of 0.05. Ordinary least squares (OLS) regression analyses were performed to determine the extent to which human V1 and LGN volumes could be predicted from nonhuman scaling relationships. In addition, reduced major axis (RMA) regression analyses were performed to compare the slopes of different taxonomic groups, and to determine intraspecific scaling patterns within the *H. sapiens* and *P. troglodytes* samples.

In accordance with previous studies (Frahm et al., 1984; Holloway, 1997; Bush and Allman, 2004), regressions for higher-level taxonomic groups were estimated (e.g., “primates”) and the difference between human observed and predicted values were computed for several taxonomic levels. However, to compare human brain organization with that of closely-related species, the

sampling strategy focused on hominoids. In addition, comparisons of human predictions drawn from different reference samples provide an estimate of reliability and sensitivity of the regression equation to idiosyncrasies of the sampling design (Holloway and Post, 1982).

In the analyses, data from the hominoid specimens and the *Macaca fascicularis* specimen were included along with previously published non-hominoid primate volumetric data for LGN (Stephan et al., 1984), V1 (Frahm et al., 1984), and brain weight data (Stephan et al., 1988; see [Supplementary Online Materials Table 2](#)). The human V1 and brain volumes were published previously (Amunts et al., 2007a). In all statistical analyses, left V1 and left LGN volumes were doubled to estimate the total (left plus right hemisphere) volumes of V1 and LGN for each specimen because the volumes of V1 (Amunts et al., 2007a) and LGN (H. Frahm, unpublished observation) apparently do not exhibit major asymmetries. Because autocorrelation is a potential source of error in cases where a part of a structure is regressed against a whole (Deacon, 1990), the value of the part was subtracted from the value of the whole and data were log₁₀ transformed. Regression equations were calculated from species mean values but human values were excluded. Human mean values and individual values were plotted as points on graphs for visualization of intraspecific variation.

The aforementioned regressions used to predict human values, based on standard “tip” data which does not take into account phylogeny (TIP), were compared to phylogenetic independent contrasts (PIC) ordinary least squares regressions. The phylogenetic independent contrasts method uses information about phylogenetic relationships (topology and branch length) to calculate contrasts from pairs of monophyletic groups (species and clades) joined at nodes (Felsenstein, 1985). In this way, PIC can be used to adjust data for differences in scaling relationships between phylogenetic groups. Contrasts were calculated in the PDAP:PDTree module version 1.07 (Midford et al., 2005) of Mesquite software version 2.0 (Maddison and Maddison, 2006) to create OLS slopes while controlling for phylogenetic relatedness. The primate phylogeny and branch lengths were taken from Purvis (1995), and all polytomies were treated as soft polytomies for determining degrees of freedom (Purvis and Garland, 1993; Garland and Diaz-Uriarte, 1999). The PIC regression intercept was mapped back onto the original data space to calculate predictions of human values, using y-intercept values calculated in PDAP (Garland et al., 1992; Garland and Ives, 2000). Two types of OLS PIC regressions were calculated: generic (G-PIC) regressions and *Homo sapiens*-specific (H-PIC) regressions (Garland and Ives, 2000; Ross et al., 2004). Methodologically, these differ only in the position of the root node; this results in a different y-intercept but does not alter the slope. For the G-PIC regressions, the primate phylogeny used omits

Table 2

Chord distances between external brain landmarks, and ratios between chord distances for comparison with volume ratios

species	code	Chords (mm)		Chord ratios	Volume ratios	
		OP-LS	OP-FP	OP-LS/ OP-FP	V1/ Brain	V1/ Neocortex
<i>Pan troglodytes</i>	Bathsheba 4/97	27.14	78.99	34.36%	2.71%	3.58%
<i>Pan paniscus</i>	Zahlia 1/97	30.08	74.23	40.52%	3.64%	5.30%
<i>Pongo</i>	Harry 2/97	17.60	82.40	21.36%	1.96%	2.78%
<i>pygmaeus</i>						
<i>Pongo</i>	Briggs 5/97	22.20	78.18	28.39%	2.82%	3.91%
<i>pygmaeus</i>						
<i>Hylobates lar</i>	Disco 3/97	22.74	52.99	42.92%	3.96%	5.95%

For all correlations between chord ratios and volume ratios, Spearman's rho $r = 0.9$, $p < 0.05$

“OP” occipital pole, “LS” lunate sulcus, “FP” frontal pole.

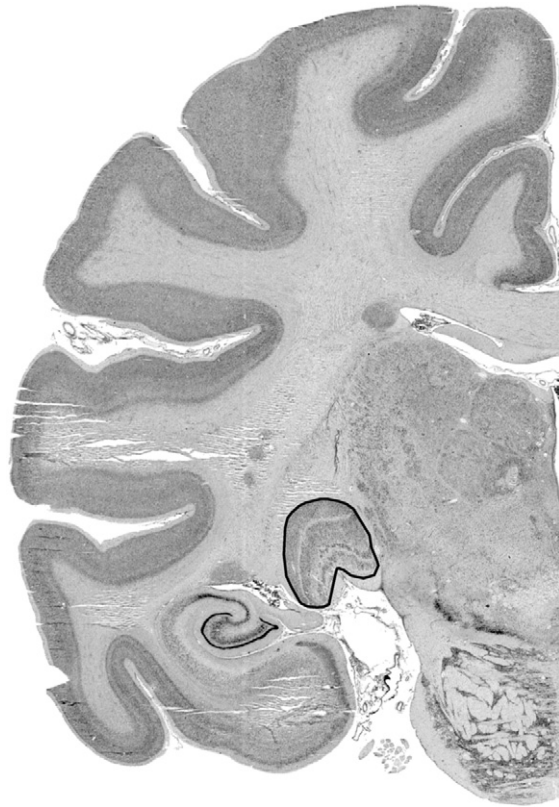


Fig. 2. *Hylobates lar* Nissl-stained left coronal section through the LGN (outlined).

H. sapiens and the root node is that which links the strepsirrhine and haplorrhine primates; these regressions are best suited for comparison of scaling coefficients to the standard OLS regressions. For the H-IC regression, the root was moved to the branch leading to *H. sapiens*, and then *H. sapiens* was pruned from the tree as described by Garland and Ives (2000), for predicting the values of the unmeasured species. The H-PIC procedure displaces the intercept of the regressions towards the new root of the tree, that is, towards the position that would be occupied by *H. sapiens*.

The percent difference between observed (O) and predicted (P) human values (O/P % difference) was calculated from the nonhuman regression equations for V1 as functions of brain minus V1 volume, LGN as functions of brain minus LGN volume, and V1 as a function of LGN volume (Holloway, 1997, 2002).

The volume ratios of V1/brain, V1/neocortex and OP-LS/OP-FP chord ratios were analyzed with nonparametric Spearman's rank correlations. The calculation of multiple correlations was not corrected because of the exploratory nature of the present study.

Results

Absolute visual structure volumes

Among hominoids, absolute left V1 volume is greatest in humans, and lowest in the hylobatids, which are phylogenetically most distant from humans. The mean and inter-individual variability in the human sample reported here ($n = 10$; mean = 7.62 cm^3 ; CV = 0.23) is similar to that reported in a previous study ($n = 9$; mean = 7.91 cm^3 ; CV = 0.19; Gilissen and Zilles, 1995). The range of variation in human V1 volume is similar to that for chimpanzee V1 volume (*Pan troglodytes* $n = 7$; mean = 4.13 cm^3 ; CV = 0.20) and for human LGN volume (*H. sapiens* $n = 10$; 0.17 cm^3 ; CV = 0.27).

The largest nonhuman V1 volume is that of a bonobo (*Pan paniscus*), near the human mean. A nonparametric test for two independent samples found that the *H. sapiens* sample ($n = 10$) was significantly different from the *Pan troglodytes* sample ($n = 7$; $p < 0.001$; Mann-Whitney $U = 0.000$). No significant difference was found between humans and bonobos, or between chimpanzees and bonobos, but this may be due to small sample size. Absolute left LGN volume followed a pattern similar to V1 volume, with humans having the largest mean value and gibbons having the lowest mean value. However, there was more intraspecific variability and inter-specific overlap in the ranges of individual values than for V1, and a nonparametric Kruskal-Wallis test failed to detect significant differences between species.

Visual structure volume ratios

The ratio of V1 to LGN volume is greatest in humans and lowest in hylobatids (Fig. 3). However, the largest individual V1/LGN ratio belongs to a chimpanzee (*Pan troglodytes*). A Mann-Whitney U test did not find *H. sapiens* to be significantly different from *Pan troglodytes* in V1/LGN ratio, and the *Homo-Pan* clade differed significantly from all other hominoids treated as a single paraphyletic group ($p < 0.01$; Mann-Whitney $U = 15$).

The ratio of V1 to neocortex volume follows an inverse trend: it is lowest in humans and highest in *Hylobates* (Supplementary Online Materials Table 3). All non-human great ape ranges overlap with each other, but not with the humans and hylobatids. The ratio of V1 to brain volume is similar to the V1/neocortex volume ratio. Notably, the bonobo ratios cover a large range, with one specimen falling within the range of the other ape values, and overlapping in its 95% confidence interval with the hylobatids. Orangutans (*Pongo pygmaeus*) have the lowest V1/brain ratio among apes and do not overlap with the human values. The ratio of LGN to brain volume is highest in *Hylobates* and lowest in *Homo*. All non-human great ape values overlap, but not with the humans and hylobatids.

Visual structure volume regressions

Here we focus on the TIP data for two reasons: first, because the G-PIC regressions tended to fall within the 95% confidence intervals of the TIP regressions, and second because in some cases PIC regressions were not significant whereas TIP regressions were significant. Given the TIP data, a typical primate brain of human size would be expected to have a V1 volume 116% larger than the actual average human V1. These values are higher when comparing LGN volume to brain volume. A typical primate brain of human size would be expected to have an LGN volume that is 171% larger than the actual average human LGN.

The smallest monophyletic group (except humans) that had statistically significant regressions from the TIP data was "nonhuman hominoids" (Figs. 4–6). For PIC scaling relationships (which were drawn from even fewer ($n-1$) data points), the only significant regression for hominoids was for LGN/brain-LGN.

The percent difference between human observed and predicted values (O/P % difference) were calculated for V1 volumes scaled to LGN volumes to indicate whether humans have an unusual relationship between major cortical and subcortical visual structures. The difference between observed and predicted human values was relatively low regardless of taxonomic level of comparison. Values ranged from humans having a V1 volume that is 33% larger than expected for a primate of similar LGN size, to humans having a V1 volume that is 18% larger than expected for a nonhuman hominoid with the same LGN volume. Therefore, although the volume of V1 is greater than expected in humans given the amount of LGN input,

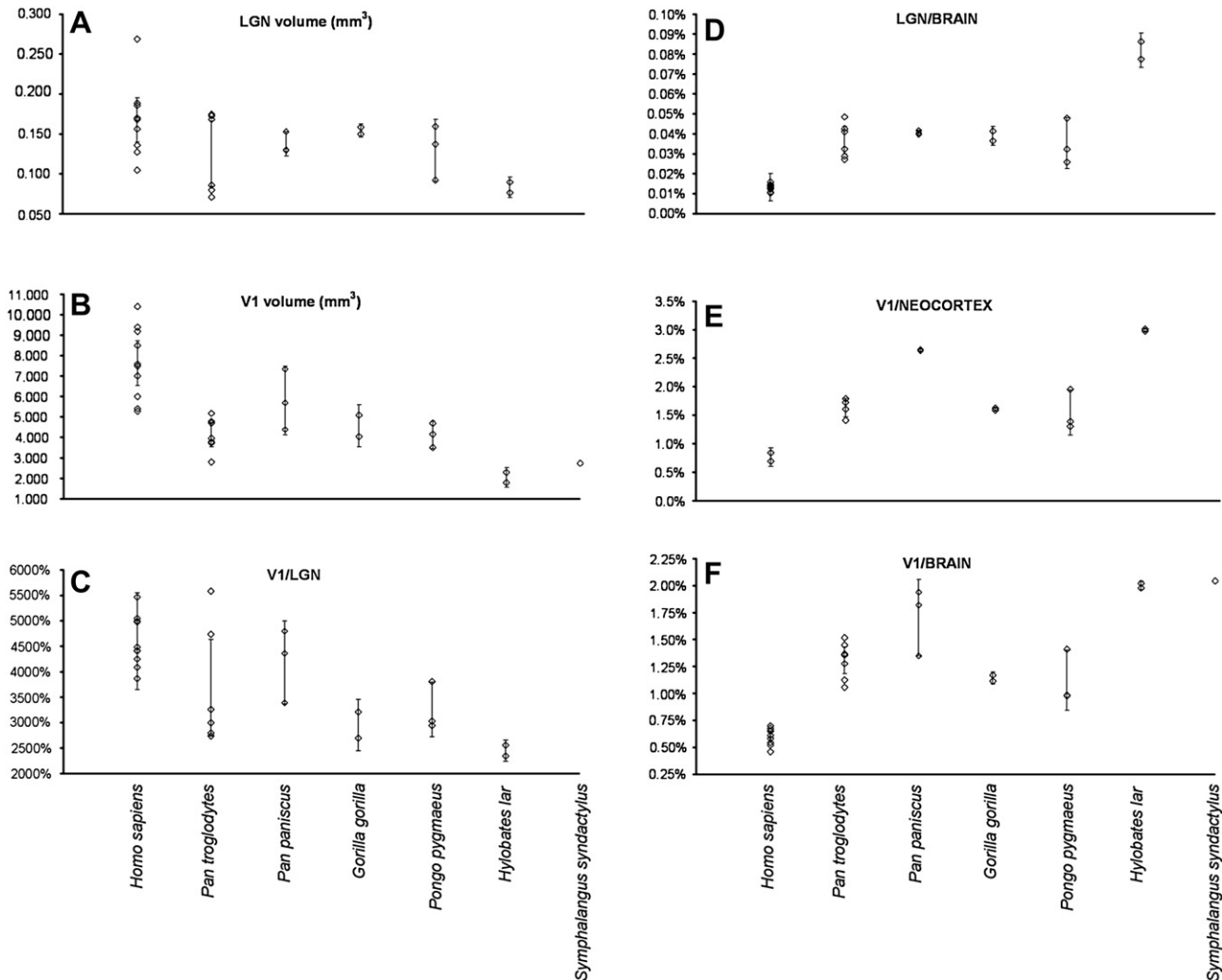


Fig. 3. Individual specimen LGN volumes (A), V1 volumes (B), V1/LGN volume ratios (C), LGN/brain volume ratios (D), V1/neocortex volume ratios (E), and V1/brain volume ratios (F). Vertical bars indicate 95% confidence intervals of species mean values. Note that only V1 volume and V1/brain volume ratio data are available for *Symphalangus syndactylus*.

the magnitude of the percent difference in observed and predicted values does not indicate that human V1 volume is particularly large relative to a nonhuman primate or hominoid of similar LGN size.

The H-PIC percent difference between observed and predicted human values was calculated to determine whether phylogenetically-informed predictions for humans produced values that are more closely approximated to observed data. Surprisingly, the H-PIC % difference O/P *H. sapiens* values was in some cases actually higher than the TIP % difference O/P *H. sapiens*: this was the case for LGN volume predicted for a nonhuman hominoid of similar brain size, and V1 and LGN from brain size predictions based on published catarrhine regressions.

Differences between hominoid and cercopithecoid regressions The G-PIC OLS regressions predicting V1 and LGN volume from brain volume for catarrhines fall outside the 95% confidence interval of the TIP OLS regression equations, yielding very different percent differences between observed and predicted human values (Figs. 5,6, Supplementary Online Materials Table 4). To further explore the taxonomic differences in scaling, RMA regressions were calculated for the hominoids and their sister taxon, the cercopithecoids, and then compared to higher taxonomic levels (Fig. 7, Supplementary Online Materials Table 5). The RMA equations for V1 as a function of brain minus V1, and of LGN as a function of brain minus LGN, show a decrease in slope from higher to lower

taxonomic level. However, for both RMA regressions, the cercopithecoid slope is steeper than that of any other group, indicating that both visual brain structure volumes increase rapidly as a function of brain size (Fig. 7). The RMA equation for V1 as a function of LGN volume increases from higher to lower taxonomic level. However, for this RMA regression, the cercopithecoid slope is flatter than that of any other group, indicating that V1 increases slowly as a function of LGN size in this group. Using a resampling test in SMATR (Warton et al., 2006), cercopithecoids and hominoids were found to be significantly different in the RMA slopes for V1 as a function of brain minus V1 ($t = 5.956$, $p = 0.011$), for LGN as a function of brain minus LGN ($t = 8.590$, $p = 0.007$), and for V1 as a function of LGN ($t = 4.035$, $p = 0.030$).

In the plot of individual and species mean values for V1 volume as a function of brain minus V1 volume, the human mean value is bounded by the hominoid (including human) OLS regression 95% prediction intervals, and most human values fall between the regression and the lower prediction interval (Fig. 8A). The bonobo values are very high, with one value falling above the prediction interval, and the mean near the upper prediction interval. Other ape values aggregate around the regression.

In the plot of individual and species mean values for LGN volume as a function of brain minus LGN volume, the human mean value is bounded by the 95% prediction intervals, and most human

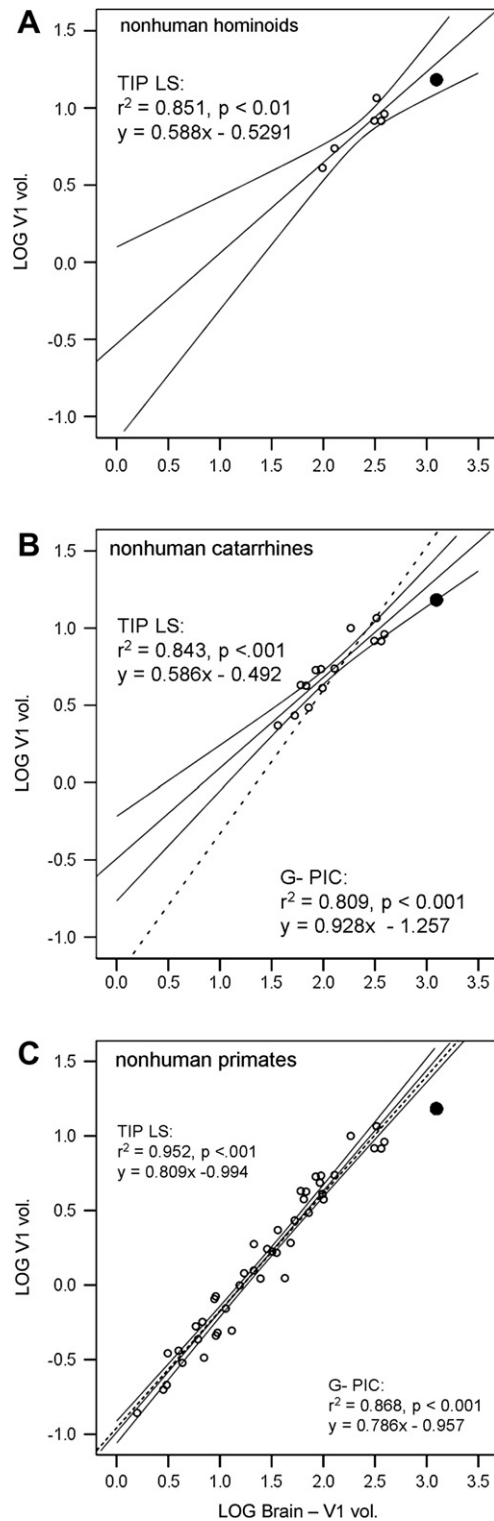


Fig. 4. TIP OLS regression lines and 95% confidence intervals for predicting human values (solid lines). Where significant, G-PIC OLS regressions are shown (dotted lines). (A) V1 as a function of brain-V1: nonhuman hominoid, (B) nonhuman catarrhine, and (C) nonhuman primate. Human values are plotted as large filled circles.

values fall between the regression line and the lower 95% prediction interval—although one value fell just below the upper prediction interval, and two values fell below the lower prediction interval (Fig. 8B). A similar spread of values was found for *Pan*

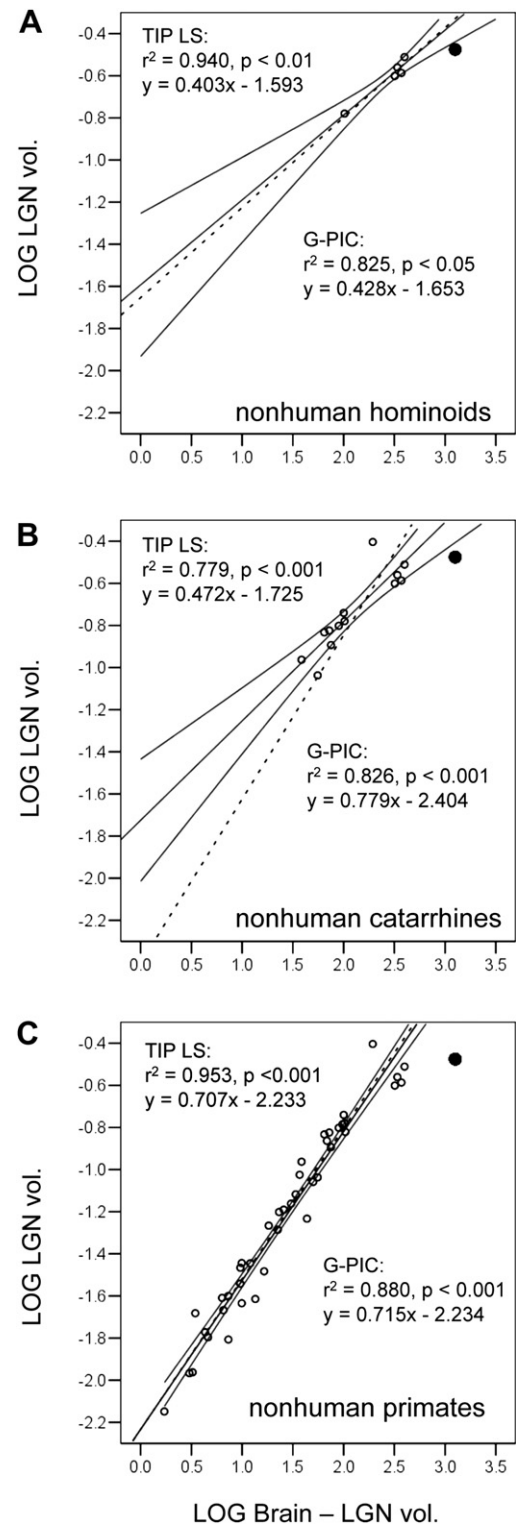


Fig. 5. TIP LS regression lines and 95% confidence intervals for predicting human values are shown (solid lines). Where significant, G-PIC LS regressions are shown (dotted lines). (A) LGN as a function of brain-LGN: nonhuman hominoid, (B) nonhuman catarrhine, and (C) nonhuman primate. Human values are plotted as large filled circles.

troglodytes, for which half lie above the upper prediction interval, and half lie below the lower prediction interval. Similarly, the orangutan values showed a large range of variation, with one below the lower prediction interval. In contrast to their especially large V1

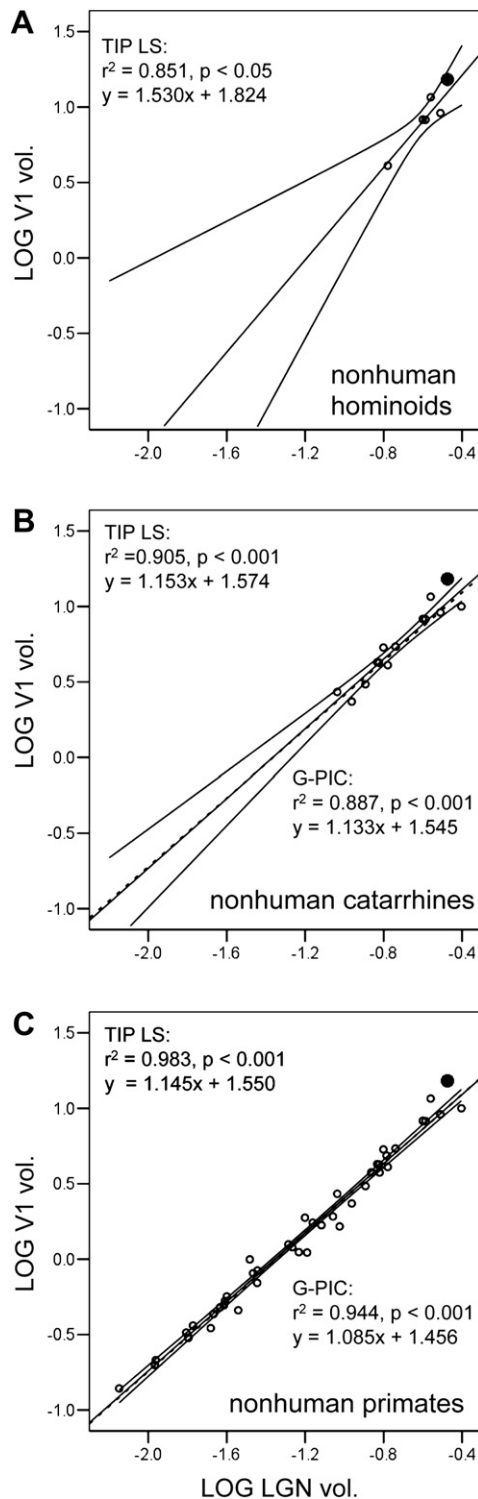


Fig. 6. TIP LS regression lines and 95% confidence intervals for predicting human values are shown (solid lines). Where significant, G-PIC LS regressions are shown (dotted lines). (A) V1 as a function of LGN: nonhuman hominoid, (B) nonhuman catarrhine, and (C) nonhuman primate. Human values are plotted as large filled circles.

volumes, bonobos do not have especially large LGN volumes. All bonobo, gorilla, and gibbon specimen values fell within the prediction interval brackets.

All hominoid species mean values for V1 volume as a function of LGN volume are bounded by the 95% prediction intervals (Fig. 8C).

Most values fall within the prediction intervals, with the exception of one human and two chimpanzee values which fall above the upper 95% prediction interval.

Human and chimpanzee intraspecific RMA regression slopes were not significantly different, although the intercepts differed significantly as indicated by pair-wise Wald tests for shifts in elevation: V1 as a function of brain minus V1 (Wald statistic = 21.466, $p < 0.001$), LGN as a function of brain minus LGN (Wald statistic = 67.707, $p < 0.001$), V1 as a function of LGN (Wald statistic = 6.638, $p < 0.05$; See supplementary Table 5 for further details).

Lunate sulcus position in relation to V1 volume

A nonparametric test found a correlation between the five ape specimens' OP-LS/OP-FP chord ratios paired with the V1/BRAIN volume ratios, and for the chord ratios paired with the V1/NEOCORTEX volume ratios (in all cases, Spearman's $\rho = 0.9$, $p < 0.05$; Table 2). This indicates that ape brains with a more anteriorly-positioned lunate sulcus also have larger relative V1 volumes (Fig. 9). However, because the correlation is less than 1.0, 19% of the variability in V1 volume is not explained by the linear distances between the landmarks LS, OP and FP. The remaining variability could be attributed either to differences in the proportion of V1 on the medial surface or in sulci, or to differences in the shape of the hemispheres and the shape of the lunate sulcus.

Discussion

Human predictions

The main purpose of this study was to determine whether brain reorganization in *H. sapiens* is manifest in an unusually small V1 volume for its brain size. Because the LGN is the primary source of inputs to V1, the relationship of LGN size to brain size, and V1 size to LGN size, were also considered to provide a wider scope within which to interpret the findings. Our results demonstrate that human V1 and LGN volumes are smaller than expected for a primate of similar brain volume. Specifically, the H-PIC prediction indicates that the volume of V1 is 95% smaller than predicted for a nonhuman primate of similar brain volume, indicating that, even after accounting for phylogenetic trends within primates, human have a significantly reduced V1 volume.

Consistent with the present results, other researchers have reported smaller than expected V1 and LGN volumes in humans (e.g., Frahm et al., 1984; Holloway, 1997). However, using the same primate dataset, Conroy and Smith (2007) calculated that the observed human visual cortex volume was just 18% less than predicted based on an independent contrasts regression. The present study's data set has a better representation of hominoid species, which were derived from larger-sized samples, plus one additional catarrhine species; this is in contrast with the Stephan et al. (1981) data-set used in previous studies. Also, here volumes for V1 grey matter, rather than grey and white matter, were used because these could be estimated more directly (Frahm et al., 1984). Thus the results of the present study are based on a more representative sample than used previously.

Hominoid diversity

In contrast to previous studies in which all species were represented by limited sample sizes, the current study evaluated the degree of intraspecific variability by including more hominoid individuals. Our new data demonstrate that hominoid values are more variable than previously appreciated. Given this range,

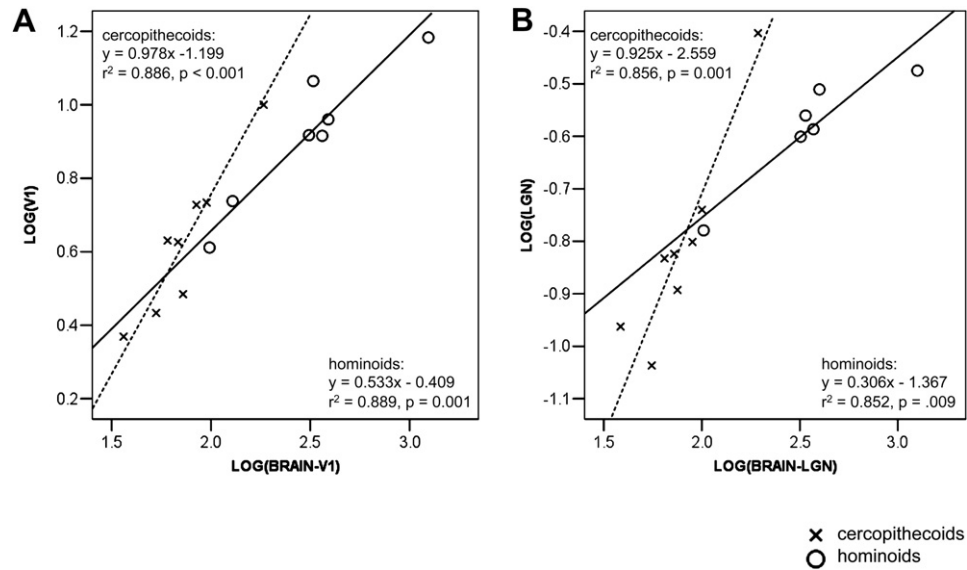


Fig. 7. Cercopithecoid and hominoid RMA regressions of TIP data of (A) V1 as a function of brain-V1 and (B) LGN as a function of brain-LGN.

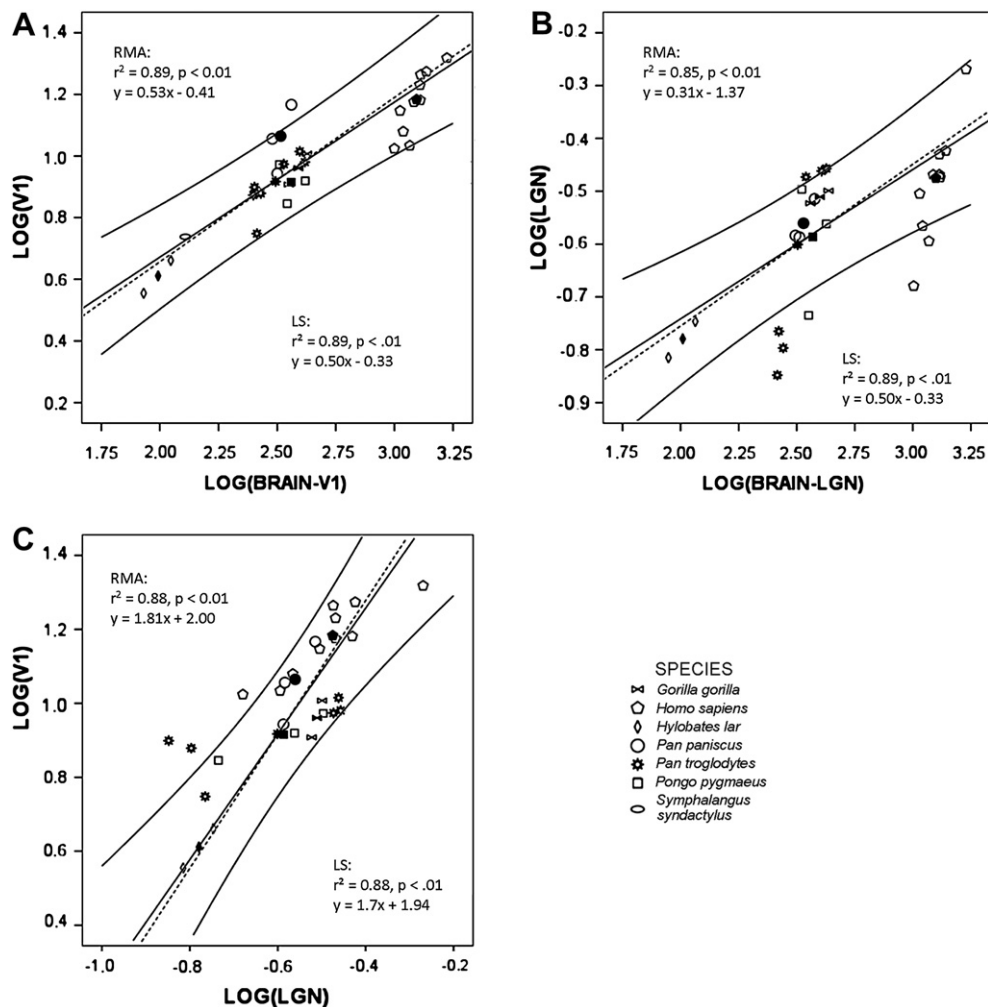


Fig. 8. Hominoid regressions of TIP data for (A) V1 volume as a function of brain-V1 volume, (B) LGN volume as a function of brain-LGN volume, and (C) V1 volume as a function of LGN volume. The solid lines are the OLS regressions of the species means, with upper and lower 95% prediction intervals. The dashed lines are the RMA regressions of the species means. The human mean was included in the calculation of the regression equations. Both the individual values (empty symbols) and the species means (filled symbols) are plotted.

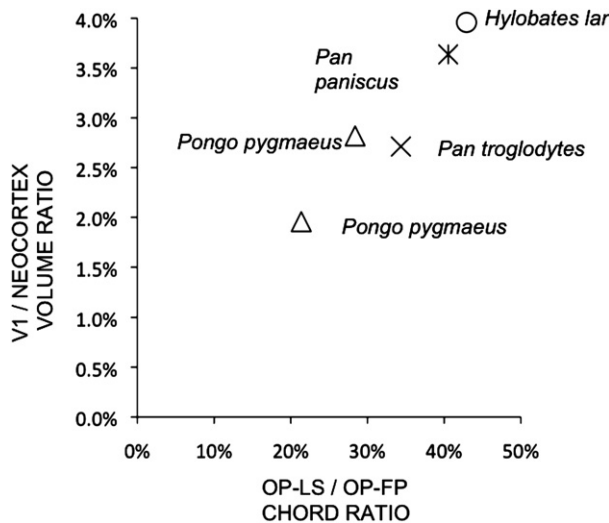


Fig. 9. Scatter-plot showing the relationship between the occipital pole and lunate sulcus (OP-LS) as a fraction of occipital pole to frontal pole (OP-FP) chord ratios and the V1/neocortex volume ratios for 5 ape individuals.

humans overlapped with nonhuman hominoids in the absolute volumes of V1 and LGN. Also, *Pan paniscus* V1 volumes appear to be particularly large, both absolutely and when compared to brain and LGN volumes.

Variability in panin V1 organization

The bonobo (*Pan paniscus*) V1 volumes stand out as being absolutely large for great apes, and relatively large for hominoids. The degree to which the human V1 volume deviates from the nonhuman hominoid prediction is similar to the degree from which the bonobo V1 volume deviates from the hominoid prediction, but in a different direction. In terms of absolute V1 and LGN volumes, bonobos overlap with the human range, although it should be noted that the current bonobo sample size is rather limited and observations made here may be indicative of a larger trend and warrant further investigation. Interestingly, a study on a subset of these specimens found that the single bonobo investigated was most similar to humans in having a low volume fraction of cell bodies in V1 (GLI; de Sousa et al., 2009). This could indicate that the increase in V1 volume in humans and bonobos is accompanied by increased space for neuropil and potentially more interneuronal connections.

Cercopithecoïd–hominoid differences

The trend is that the lower the taxonomic level of the group, the less steep the slope, as reported in previous studies (Clutton-Brock and Harvey, 1979; Pagel and Harvey, 1988, 1989; Stephan et al., 1988). This is seen for the RMA regressions of V1 as a function of brain minus V1, and LGN as a function of brain minus LGN. The taxon-level effect has been ascribed evolutionary significance, such as differential selection acting in body size versus brain size in closely related species (Gould, 1966; Lande, 1979). It has also been suggested that the steeper slopes of higher taxa result from linking the graded shallower slopes of lower taxa, and that the grade-level differences are due to the particular ecological conditions of that group (Pagel and Harvey, 1989). However, there are two deviations from what at first appears to be a “taxon-level effect” in the scaling of V1 and LGN volume to brain volume. First, the steepest slopes were found for the interspecific comparisons (Supplementary

Online Materials Table 5); chimpanzee LGN volume increases more rapidly than does its brain size, and humans' V1 and LGN volume increase more rapidly than does brain size (Fig. 8). The intraspecific variability in V1 volume and LGN volume may be related to variability in size of the retina; unfortunately, this cannot be tested as these data are not available.

Second, it is noteworthy that in the sister group of the hominoids, the cercopithecoïds, LGN and V1 increase rapidly in volume with increasing brain volume, as compared to other primate groups. This is in contrast to the hominoid trend, and could indicate a cercopithecoïd-specific organizational pattern of the visual system. Generally, the visual system of the two major catarrhine taxa, cercopithecoïds and hominoids, are similar, sharing primate synapomorphies such as stereoscopy, as well as a notable catarrhine synapomorphy, routine trichromatic vision (Jacobs and Deegan, 1999; Deegan and Jacobs, 2001). However, it is becoming increasingly apparent that cercopithecoïds and hominoids differ in aspects of visual system neuroanatomy, such as lamination of V1 (Preuss et al., 1999) and the percentages of GABAergic interneurons in V1 and V2 (Sherwood et al., 2007). The different patterns of scaling of visual system brain structures to brain size in cercopithecoïds and hominoids indicates that these two taxonomic groups have differences in brain organization.

Lunate sulcus position as an indicator of reorganization

Paleoneurological studies equate lunate sulcus position with the relative volume of the visual cortex, but this assumption had not been tested. The results of the present study, which are based on chord measurements from a small sample, indicate an overall relationship between relative V1 volume and lunate sulcus position. To our knowledge, this is the first test of the aforementioned assumption. Furthermore, although indirect, support comes from developmental studies suggesting that the relationship between the V1/V2 border and the lunate sulcus position is not entirely arbitrary. In spite of the reduction in V1 area in enucleated macaques, the V1/V2 border still usually occurred near a sulcus, and when it did not, a small “kink” within V2 occurred near the V1/V2 border (Dehay et al., 1996).

Although the lunate sulcus has previously been shown in chimpanzees to be a reliable delimiter of the lateral-anterior extent of V1 (Holloway et al., 2003a), the lunate sulcus position may not be enough to make generalizations about V1 volume because it does not take into account portions of V1 represented in inferior and medial aspects of the cerebrum, and does not account for potential differences in cortical thickness or the degree of cortical folding. Further, the current MRI-based study was limited to chord measurements which do not account for the overall shape of the cerebrum, although this is accounted for by using arc measurements on endocasts (e.g., Holloway, 1983). For example, it is known that among anthropoids, there is an overall increase in the degree of cortical folding corresponding to increasing brain size (Zilles et al., 1989). Overall, gyrification is higher in humans than in great apes, but in the occipital lobe humans and great apes have a similar degree of gyrification (Armstrong et al., 1991). Although the aforementioned factors cannot be addressed by the current dataset, they certainly warrant further investigation.

If, as suggested here, lunate sulcus position is related to V1 volume, then a posterior lunate sulcus in early, small-brained hominins would indicate V1 volumes smaller than those of great apes. Significant variation in V1 volume within humans related to age (Leuba and Kraftsik, 1994), sex (Amunts et al., 2007a) and population (Klemp et al., 1994), as well as intraspecific and intra-individual differences in lunate sulcus position in chimpanzees

(Holloway et al., 2003a), have not been linked to differences in basic visual functions. It is important to consider that volume is only one feature of V1, and other variables not taken into account include neuron density, size, type, and connections. A reduction in V1 volume would not necessarily correspond to a reduction in total amount of space occupied by neuronal cell bodies, or in the number of neurons. For example, a study on a subset of this sample found that for a group of closely related primate species (hominoids plus *Macaca fascicularis*), greater neuron volume densities were found to correspond to smaller V1 volumes, and not overall brain size (de Sousa et al., 2009).

Evidence of intraspecific variability in relative V1 volume presented here, along with intraspecific variability in chimpanzee lunate sulcus position (Holloway et al., 2003b), indicates that some fossil specimens like Stw 505 could be at one end of the distribution. However, the differences observed here between chimpanzees and bonobos in absolute V1 volume, and in microanatomical organization, indicate that species-level variability in brain organization, irrespective of brain size, may also have existed among early hominins. Dart (1925) proposed that the expansion of parieto-occipito-temporal association areas without a change in brain volume led to a reduction in primary visual cortex volume in Taung. Holloway (1966, 1968) linked the expansion of posterior parietal association areas inferred from early hominin endocasts to functions such as advanced communication, tool use and tool-making, social complexity, and long term memory. Further, Bruner and colleagues (Bruner et al., 2003; Bruner, 2004, 2007) related functionally-relevant brain reorganization and parietal lobe expansion during the Pleistocene. Globularization of the *Homo sapiens* neurocranium reflects parietal lobe expansion, in contrast to that of earlier *Homo*. Such changes have been explained in relation to increased tool-making sophistication and language abilities (Bruner et al., 2003; Bruner, 2004, 2007).

Conclusions

This study provides evidence for gross-level brain reorganization within the hominoids and catarrhines, with respect to the relative size of V1 and LGN. V1 volumes are quite variable within and among hominoid species, supporting earlier findings that, within the hominin lineage, V1 volume is smaller than expected for a primate of similar brain size. Furthermore, V1 volume is predictable from lunate sulcus position in apes. Therefore, evidence of posteriorly-positioned lunate sulci in early fossil hominins suggests that V1 reduction began early in hominin evolution. Decreases in V1 volume may correspond to increases in the volumes of adjacent visual areas. The functional consequences of such a reconfiguration of cortical area volumes are unknown, although it has been hypothesized that the “reduced” V1 of *Australopithecus africanus* would remain sufficient for primary sensory function, while enabling reallocation of surrounding tissues to more specialized functions (Holloway et al., 2001).

These findings support the notion that, if brain sulci are identifiable on fossil endocasts, it might be possible to make inferences about the brain organization of extinct species. Human paleoneurology would benefit from further information about whether a scaling relationship between lunate sulcus position and V1 volume can be used to make quantitative predictions about V1 volume in fossil hominin specimens. Further histometric work is required to determine whether the lunate sulcus position correlates with the volumes of higher-order parietal association areas. Finally, similar methods could be applied to other sulci to investigate their paleoneurological relevance as landmarks for cytoarchitectonic borders.

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Appendix. Supplementary data

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhevol.2009.11.011

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